Monthly Reflection Report

AI/ML Internship — Month 1 Summary

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Overview

This past month has been the real starting point of my internship. I went from setting up a clean working environment to actually cleaning data, building visualizations, and putting together a working dashboard. I have also work with different tools to clean data and create different types of visuals to understand data and its trends.

Week 1

Orientation and Setup:

The first week was mostly about getting organized. I set up the internship folder structure, created the GitHub repository, and wrote my introduction along with a one-page document on my learning goals. I also spent time on revising up on the core concepts of AI/ML — AI, machine learning, deep learning, generative AI, LLMs, RAG, vector databases, and prompt engineering, and write it down in my own words helped me realize how much I already knew and where the gaps still were.

Week 2

Python Practice and the Data Health Checker

In second week of my internship I revised the core concepts Python — lists, dictionaries, functions, loops, file handling, and exception handling , and create a practice python notebook on these concepts. In this week I built a Data Health Checker utility that takes any CSV file and reports its shape, missing values, duplicates, data types, and a basic statistical summary, along with suggested cleaning actions. For this work i use Pandas library .

Week 3

Data Cleaning and Exploratory Data Analysis

In this week I work on real-time dataset and learn how to clean dataset and transform it into useful information by doing EDA on it. I worked with the Netflix dataset, cleaned it using Pandas — removing duplicates, handling missing values, fixing column names, and correcting data types — and ran into a couple of real bugs along the way. One was a logic error where duration values that had landed in the wrong column were getting dropped instead of moved to where they belonged. The other was a notebook that kept giving me a stale state, so my fixes weren't actually showing up until I re-ran things properly. Once both were sorted, I ended up with a fully clean dataset of about 8,709 rows and zero missing values. I wrote out 10 insights in plain language and put together the EDA report.

Week 4 —

Visualization and the Streamlit Dashboard

Using the cleaned dataset from Week 3, I built several visualizations covering everything from bar and line charts to a correlation heatmap, each with a business question behind it, a reason for choosing that chart, and an insight drawn from it. Then I built the Public Data Insights Dashboard in Streamlit. This went through several rounds of changes — I moved the filters from the sidebar to a horizontal layout at the top, swapped out a duller color palette for a more vibrant and professional one, added short 'what this shows' notes under each chart, and rewrote the Key Insights section so it actually explained business significance instead of repeating what the chart already showed. I also added a Recommendations section with priority-tiered cards built from live dataset statistics.

What I Learned

- Clean code and clear documentation matter just as much as getting the output right — a good README and sensible commit messages are part of the deliverable, not an afterthought.
- Debugging isn't always about finding one bug. In Week 3, I had two separate issues compounding each other, and I only found the second one after fixing the first.

- Writing insights in plain, professional language is harder than it sounds. It's easy to either sound too casual or to swing the other way and sound artificial. Finding the middle ground took a few tries.
- A dashboard isn't just charts on a page — filters, layout, color choices, and the way insights are worded all affect whether a non-technical person can actually use it.
- Consistency in small things, like keeping units the same across charts, is easy to overlook until someone else points it out.

Challenges I Faced

The biggest challenge was probably the data cleaning bug in Week 3—it took a while to realize that the notebook wasn't reflecting my actual fix because of a stale execution state, which was confusing before I figured out what was going on. I also encountered an issue where some values were misplaced after the cleaning process, and I had to carefully identify the cause and correct them. Beyond that, balancing all the moving pieces of the dashboard—layout, colors, filters, and documentation—took more iteration than I expected. I also had to unlearn the habit of writing documentation that sounded generic and instead make it more specific, technical, and written in my own voice.

Feedback

Suleman Sir feedback across these weeks has been consistent in pushing me toward clarity, clearer documentation, clearer commit history, and clearer storytelling in my analysis. Each round of feedback led to concrete changes: restructuring my README, rewriting insights to focus on business relevance, and now going back through my GitHub repository to make sure every assignment from Week 1 onward is properly organized and complete.

Closing Thoughts

A month in, I feel like I've moved from just learning Python syntax to actually thinking like a data analyst — asking what a chart is for before I build it, and what a cleaning step is fixing before I run it. There's still a lot to improve, especially around documentation habits, but the direction feels right, and I'm looking forward to building on this next month.